

Module 08 [Advanced Topics: Java Programming]: (for Week 11:28-1/3/2026)

Topic 01 [java Programming: Anonymous object]

Problem Statement: Write a Java program that creates a Point object using anonymous object using the following getPoint() method.

```
Point pt=getPoint();
```

Topic 02 [java Programming: ArrayList<> object]

Problem Statement: Write a Java program performs the following operations using ArrayList Object

- i) Declare a ArrayList object ax to store flowers
- ii) Add three flowers: Rose, Lilie, Carnation, Daisie, and Tulip
- iii) Insert a new flower Peonie in the second position of ax
- iv) Delete the flower Lilie
- v) Find any flower in ax
- vi) Sort the list
- vii) Delete the list

Topic 03 [inheritance: use of super keyword]

Problem Statement: For the following Java program write

- i) to display x of class A in class B
- ii) statement to call getX() of class A in class B
- iii) to call parameterized constructor of in class B

```
class A{
    int x;
    public A(){
        x=0;
    }
    public A(int x){
        this.x=x;
    }

    public int getX(){
        return(x+10);
    }
}

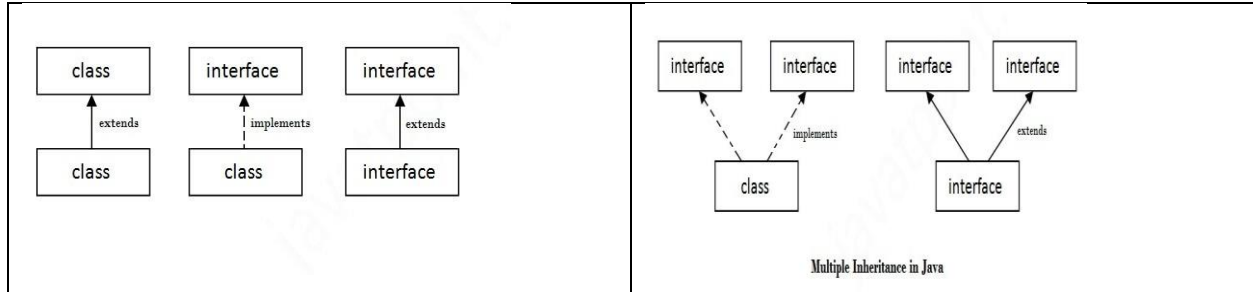
class B extends A{
    int x=20;
    public int getX(){
        return(x+10);
    }
}

public class First{
    public static void main(String[] args) {
```

<pre> } } </pre>	
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Topic 04 [inheritance]

Problem Statement: Implement the following relation diagram into java program



Topic 05 [Singleton class]

Problem Statement: Write a Java program to create and operate a singleton class.

Topic 06 [Java Multithreading]

Problem Statement: Write a program to run three applications i) Word Processor ii) Music Player and iii) Real-time clock simultaneously. Use run() method of Thread class or implements runnable interface.